Mathematical Basics.

① $x \sim p(x)$ and $f(x) \Rightarrow \underset{x\sim p(x)}{E}\left[f(x)\right] = \int_{sup(x)} p(x) f(x) dx$

② $p(x)$ and $q(x)$ are PDFs

$$D_{KL}(p \| q) = \int_{x \sim sup(p)} p(x) \underbrace{\log \frac{p(x)}{q(x)}}_{f(x)} dx = \underset{x\sim p(x)}{E}\left[\log \frac{p(x)}{q(x)}\right]$$

KL properties:

① symmetry: $D_{KL}(p \| q) \neq D_{KL}(q \| p)$

② non-negetivity $D_{KL}(p \| q) \geq 0$

$\wedge\ D_{KL}(p \| q) = 0 \Leftrightarrow p = q$

③ triangle-inequality: $D_{KL}(p \| q) + D_{KL}(q \| r)$

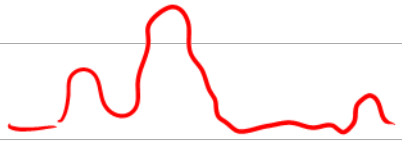
$\not\geq D_{KL}(p \| r)$

Generative Models. Given D i.i.d samples from disturbution $p_{data}(x)$, we want to approximate $p_{data}(x)$.

① Generation: Sample new x from $p(x)$

② Density Estimation: given a new x, we want to find $p(x)$

③ Finding a high level representation (latent variable)



$p(z)$ $\quad z \sim Cat[1, K]$

$+$

GMM. $\quad p(x)$ $\quad p(x|z) \sim \mathcal{N}(\mu, \Sigma)$

$\times$ too complicated $\quad \checkmark$ easy to work with

$q(z) = p(z)$

$$p(x) = \sum_z p(x, z) = \sum_z p(x|z)\, p(z)$$

$$= \sum_{i=1}^{K} \mathcal{N}(x|\mu_i, \Sigma_i)\, \pi_i$$

$p(z=i)$

$p(x|z=i)$

$z = 1, \; 2, \; \ldots, \; K$

$\downarrow \quad \downarrow \quad \downarrow$

$\Sigma_1, \mu_1 \quad \Sigma_2, \mu_2 \quad \Sigma_K, \mu_K$

new x?

$$\theta = \{\mu_i, \Sigma_i, \pi_i\}_{i=1}^{K}$$

$p(z=i) = \pi_i$

$q(z=i) = ?$

$$p(x) = \sum_z p(x,z) = \sum_z q(z) \frac{p(x,z)}{q(z)}$$

we want to maximize this!

$$= \mathop{E}_{z\sim q}\left[\frac{p(x,z)}{q(z)}\right]$$ instead we maximize this.

$$\overbrace{q(z) = p(z)}^{\text{prior}} \Rightarrow p(x) = \mathop{E}_{z\sim p(z)}\left[p(x|z)\right]$$

$$\underbrace{q(z) = p(z|x)}_{\text{posterior}} \Rightarrow p(x) = \mathop{E}_{z\sim p(z|x)}\left[\frac{p(x,z)}{p(z|x)}\right]$$

during training, we minimize $-\log p(x)$.

$$-\log p_\theta(x) = \boxed{-\log \mathop{E}_{z\sim q}\left[\frac{p_\theta(x,z)}{q(z)}\right]} \to O_1$$

$\to$ minimize this?

$$O_1 = f(E[x])$$

instead of minimizing $O_1^{(\theta)}$, we minimize:

$$O_2^{(\theta)} = -\mathop{E}_{z\sim q}\left[\log \frac{p_\theta(x,z)}{q(z)}\right] = -\mathop{E}_{z\sim q}\left[\log p_\theta(x,z)\right] \text{①}$$

$$O_2 = E[f(x)]$$

$$+\mathop{E}_{z\sim q}\left[\log q(z)\right] \text{②}$$

independent of θ!!

$$\Rightarrow O_2 \geq O_1$$

Jensen's inequality:

if f is a convex function;

$$\mathbb{E}[f(x)] \geq f(\mathbb{E}[x])$$

we want to find q, s.t. O_1 and O_2 become close!

$$D_{KL}(q(z) \,\|\, p_\theta(z|x)) = \sum_z q(z) \log \frac{q(z)}{p_\theta(z|x)}$$

we showed:

when $q(z) = p_\theta(z|x)$

then $O_1 = O_2$

and we are happy!

$$= \sum_z q(z) \log \frac{q(z)}{p_\theta(x,z)/p_\theta(x)}$$

$$= \sum_z q(z) \log \frac{q(z)\, p_\theta(x)}{p_\theta(x,z)}$$

$$= \sum_z q(z) \log p_\theta(x)$$

$$- \sum_z q(z) \log \frac{p_\theta(x,z)}{q(z)}$$

$\log p(x) = -O_1$

$$= \mathbb{E}_{z\sim q(z)} [\cancel{\log p(x)}]$$

$$- \underbrace{\mathbb{E}_{z\sim q(z)} \left[\log \frac{p_\theta(x,z)}{q(z)}\right]}_{-O_2}$$

$$= O_1 - (-O_2) = O_2 - O_1$$

☺

E-M algorithm:

① E (Expectation): use $\theta^{(t-1)}$ to set $q(z) = p_{\theta^{(t-1)}}(z|x)$

② M (Maximization): take derivative from $Q(\theta^{(t-1)})$ to find $\theta^{(t)}$:)

$$\theta^{(t-1)} = \{\Sigma_i^{(t-1)}, \mu_i^{(t-1)}, \pi_i^{(t-1)}\}$$

E-step: $q(z^{=i}) = p_{\theta^{(t-1)}}(z^{=i}|x)$

$$= \frac{p_{\theta^{(t-1)}}(x|z^{=i})\, p_{\theta^{(t-1)}}(z^{=i})}{p_{\theta^{(t-1)}}(x)}$$

$$= \frac{\mathcal{N}(x|\mu_i^{(t-1)}, \Sigma_i^{(t-1)})\, \pi_i^{(t-1)}}{\sum_{j=1}^{k} \mathcal{N}(x|\mu_j^{(t-1)}, \Sigma_j^{(t-1)})\, \pi_j^{(t-1)}}$$

M-step: $Q_q(\theta) = \underset{z\sim q}{E}\left[\log p_\theta(x,z)\right] + \cancel{\text{const}}$

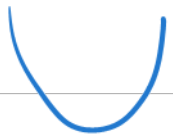
comes from E-step

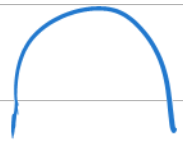
$$= \sum_{i=1}^{K} q(z=i) \log \overbrace{P_\theta(x, z=i)}^{P_\theta(x \mid z=i) \times P_\theta(z=i)}$$

$$= \sum_{i=1}^{K} \underbrace{q(z=i)}_{\gamma_i} \Big[\underbrace{\log P_\theta(x \mid z=i)} + \underbrace{\log P_\theta(z=i)}_{\log \pi_i} \Big]$$

$$\log \frac{1}{(2\pi)^{\frac{d}{2}} |\Sigma_i|^{\frac{1}{2}}} \exp\left(-\frac{1}{2}(x-\mu_i)^T \Sigma_i^{-1} (x-\mu_i)\right)$$

$$= -\frac{d}{2} \log 2\pi - \frac{1}{2} \log |\Sigma_i| - \frac{1}{2}(x-\mu_i)^T \Sigma_i^{-1} (x-\mu_i)$$

Convex 

Concave 

$f'' \geq 0$

$f'' > 0 \rightsquigarrow$ strict

$f'' \leq 0$

$f'' < 0 \rightsquigarrow$ strict

for a new sample x, and for each $i \in \{1, \dots, k\}$, we have:

$$P_\theta(z=i \mid x) = \frac{P_\theta(x \mid z=i)\, P_\theta(z=i)}{P_\theta(x)}$$

Clustering

$$= \frac{\mathcal{N}(x \mid \mu_i, \Sigma_i)\, \pi_i}{\sum_{j=1}^{K} \mathcal{N}(x \mid \mu_j, \Sigma_j)\, \pi_j}$$

$$P_\theta(x) = \sum_{i=1}^{K} p_\theta(x, z=i) = \sum p_\theta(x \mid z=i)\, P_\theta(z=i)$$

DE

$$= \sum \mathcal{N}(x \mid \mu_i, \Sigma_i)\, \pi_i$$

Generation:

① sample z from π_i $(P_\theta(z=i))$

② Sample x from $\mathcal{N}(x \mid \mu_z, \Sigma_z)$

$\downarrow$

$P_\theta(x \mid z)$

for μ_i initialization $\rightsquigarrow$ use output of another algorithm

$$\Sigma = \begin{bmatrix} Var(x_1) & Cov(x_1, x_2) & \cdots \\ & \ddots & \\ Cov(x_1, x_D) & \cdots & Var(x_D) \end{bmatrix}$$

diagonal Σ

non-diagonal